AI AUTOMATION + FINANCE

From financial variance to an explanation you can trace.

A practical prototype combining Python, SQL and an LLM commentary workflow for monthly financial review.

THE WORKED CASE

$90,000
Budget EBITDA

$34,000
Actual EBITDA

A $56,000 shortfall. What explains it?

Every result connects to source records. Every explanation remains subject to analyst review.

THE WORKFLOW

Give each part of the workflow a clear job.

The LLM receives calculated evidence and drafts commentary.

01

Validate the source data

Check unique IDs, category mappings, monthly coverage and monetary amounts.

02

Calculate with Python + SQLite

Aggregate ledger records, join budget, and calculate the EBITDA bridge in integer cents.

03

Build an evidence pack

Attach variance, transaction and management-note IDs. Preserve missing explanations.

04

Draft with an LLM

Request structured commentary, scoped citations and proposed follow-up questions.

05

Check, then review

Validate exact monetary fields and evidence IDs. An analyst reviews narrative accuracy and causes.

FINANCIAL SUMMARY

The financials behind the EBITDA bridge.

Budget versus actuals for September 2026. All amounts in USD.

Account	Budget	Actual	Variance $	Variance %
Revenue	500,000	480,000	(20,000)	(4.0%)
Cost of goods sold	200,000	204,000	4,000	2.0%
Gross profit	**300,000**	**276,000**	**(24,000)**	**(8.0%)**
Payroll	130,000	136,000	6,000	4.6%
Marketing	35,000	46,000	11,000	31.4%
Cloud services	18,000	25,200	7,200	40.0%
Rent	12,000	12,000	-	-
Professional fees	10,000	16,000	6,000	60.0%
Travel	5,000	6,800	1,800	36.0%
Operating expenses excl. D&A	**210,000**	**242,000**	**32,000**	**15.2%**
EBITDA	**90,000**	**34,000**	**(56,000)**	**(62.2%)**
Depreciation	8,000	8,000	-	-
Operating profit	**82,000**	**26,000**	**(56,000)**	**(68.3%)**

GROSS MARGIN

60.0% budget / 57.5% actual

EBITDA MARGIN

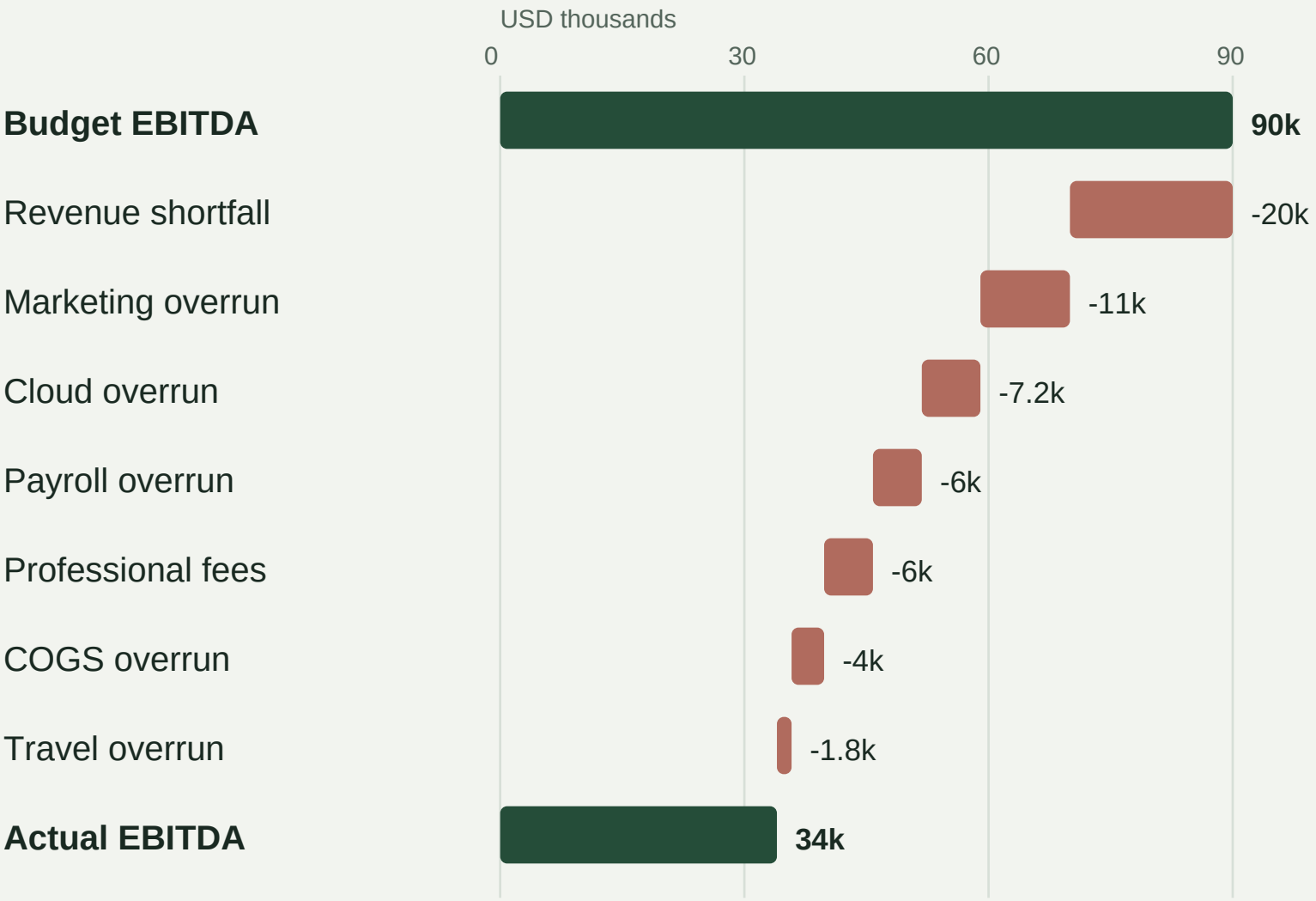
18.0% budget / 7.1% actual

Variance = actual minus budget. Higher costs and lower revenue are unfavorable. Operating expenses exclude COGS and depreciation.

CALCULATED RESULTS

Reconcile the entire $56,000 EBITDA shortfall.

Every nonzero component is included, even below the review threshold.



$20,000 less revenue + $36,000 more costs.
EBITDA margin: 18.0% budget to 7.1% actual.

THE LLM ANGLE

An explanation should show what is still unknown.

Example: cloud services cost $25,200 versus an $18,000 budget.

+$7,200	+40.0%
Expense variance	Versus budget

WHAT THE SOURCE SUPPORTS

The invoice agrees to the ledger. A workload and pricing breakdown has not been supplied.

PRE-GENERATED CHATGPT SAMPLE DRAFT

"Cloud services exceed budget. The management note confirms invoice-to-ledger agreement but provides no workload or pricing breakdown. The cause of the overrun remains unverified."

VAR-2026-09-cloud / NOTE-SEP-CLOUD

Proposed follow-up: obtain usage, unit pricing and allocation details.

CONTROLS + ANALYST JUDGMENT

Structured output helps. Evidence review still matters.

The validator checks a response contract; it cannot establish narrative truth.

CHECKED BY CODE	REVIEWED BY AN ANALYST
Exact monetary fields	Whether citations support a claim
Complete material-category coverage	Whether the cause is established
Valid evidence IDs and scope	Any numbers written in the prose
Duplicate and missing source records	Forecast assumptions and actions

19 automated checks passed

Financial reconciliation, source integrity, validation and mocked API success / failure cases.

Live API execution has not been tested. The included commentary is a sample draft, pending human review.

HOW TO PRESENT IT

Show the result.
Then show the evidence.

01 / OVERVIEW

Explain the $90,000 to $34,000 EBITDA bridge.

02 / EVIDENCE

Open cloud services and inspect the supporting records.

03 / LLM REVIEW

Show the draft, its citations and the unresolved cause.

04 / VALIDATION

Change a monetary field in the JSON and show its rejection.

THE PROJECT PACKAGE

Runnable Python + SQL, local dashboard, 324 synthetic ledger entries, Excel analysis, prompts and review controls.